

Staking pool attack simulation

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ABSTRACT

Proof-of-Stake blockchains rely on economic incentives to encourage validators to participate honestly in consensus. However, previous work has shown that these incentives can also enable Denial-of-Profit attacks, where an attacker does not necessarily try to break the blockchain itself, but instead targets a specific validator by reducing its rewards. Such attacks may be harmful to the victim, but they are often considered economically unattractive because the attacker also loses rewards. Existing analyses typically study these attacks in a fixed-stake validator setting. In modern Proof-of-Stake ecosystems, however, validators often operate as staking pools, and many users delegate their stake rather than validating directly. Since delegators can move stake between pools based on observed performance and expected rewards, an attack against a pool may have indirect market effects that are not captured by static models. This thesis investigates Denial-of-Profit attacks in the presence of staking pools, delegators, and market dynamics. To do so, it develops a configurable simulation framework and uses it to evaluate vote omission and vote delay attacks under Ethereum- and Cosmos-inspired reward mechanisms.

INTRODUCTION

Modern Proof-of-Stake blockchains rely on validators to propose blocks, vote, and finalize consensus. Rewards are distributed based on participation, stake, and whether votes are included in finalized blocks.

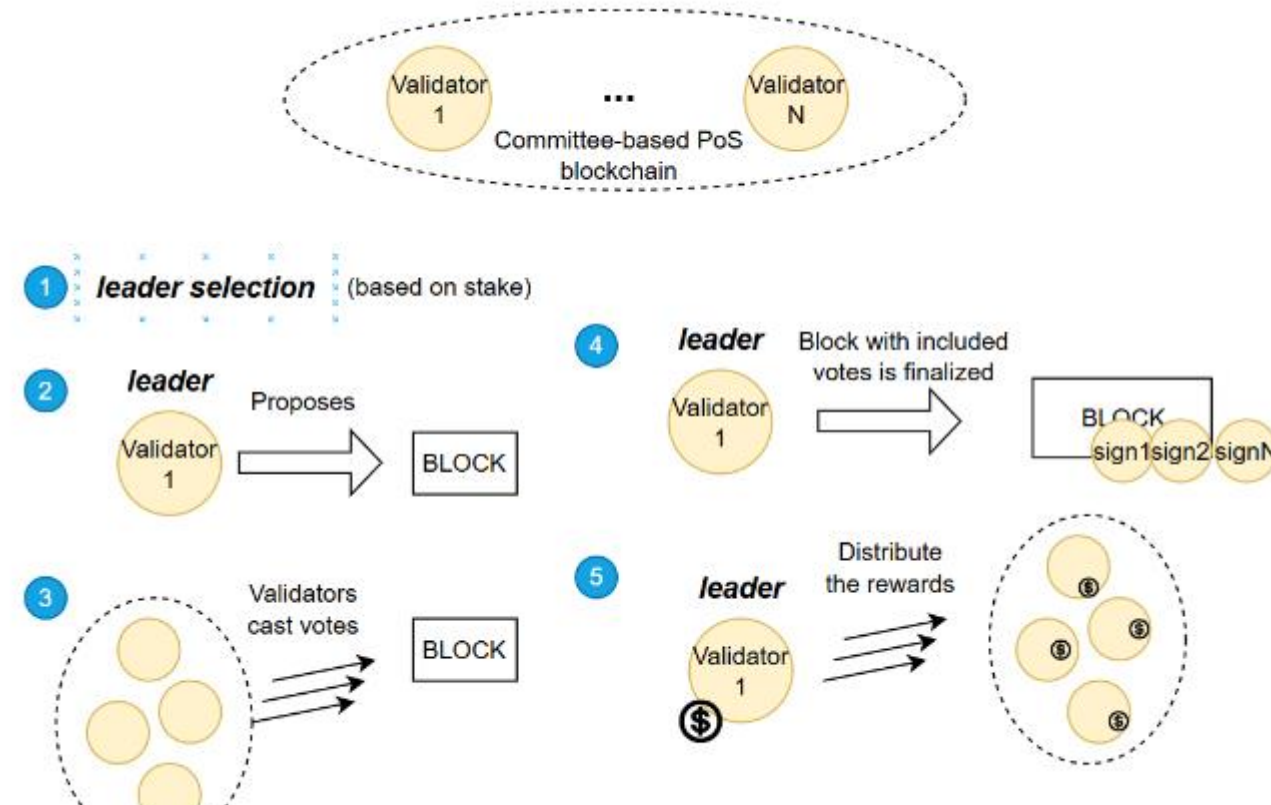


Figure 1: Normal committee-based PoS voting process

Because rewards depend on cooperation between leaders and voters, a validator can sometimes harm a specific participant without breaking the whole blockchain. These are called **Denial-of-Profit attacks**: the goal is not to stop consensus, but to reduce the reward of a chosen victim.

Two relevant attacks are considered:

- Vote omission attack**: the attacker is the leader and excludes the victim's vote from the finalized block.
- Vote delay attack**: the victim is the leader, and the attacker withholds or delays its vote for the victim's block.

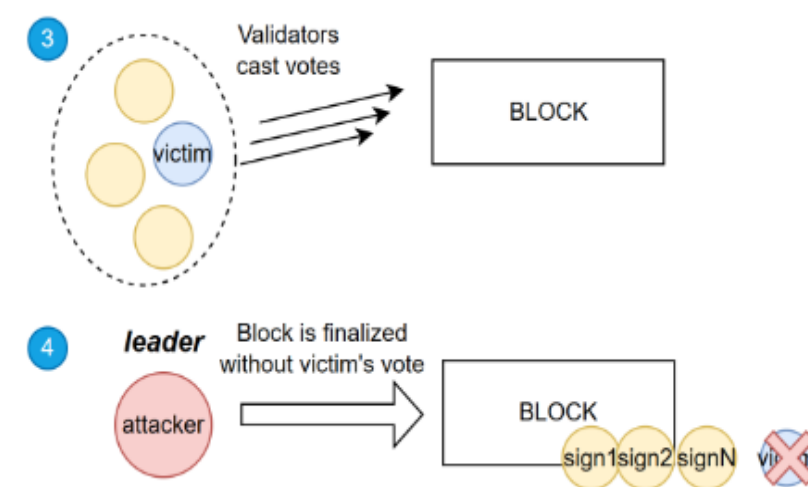


Figure 2: Vote omission attack

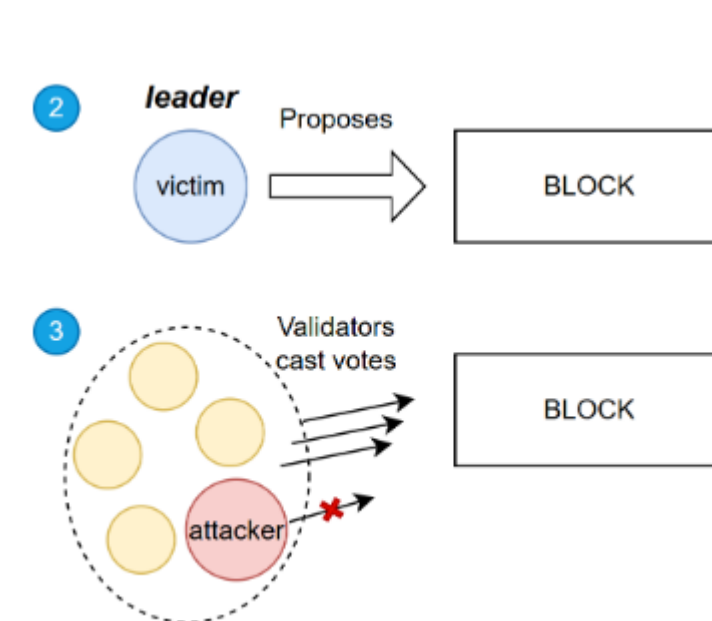


Figure 3: Vote delay attack

$$\text{Effectiveness: } \frac{\text{loss}(\text{target})}{\text{reward}(\text{target})}$$

$$\text{Attack Cost: } \frac{\text{loss}(\text{attacker})}{\text{loss}(\text{target})}$$

Modern PoS ecosystems increasingly rely on staking pools and delegation. If a pool is attacked and its performance drops, delegators may react by moving their stake to other pools. This creates a market feedback loop that is not captured by the fixed-stake analytical model.

Open problem: Can Denial-of-Profit attacks become more severe when staking pools, delegator migration, and market dynamics are taken into account?

PURPOSE OF WORK

The goal of this thesis is to study whether Denial-of-Profit attacks become more severe when they are evaluated in a staking-pool market rather than in a fixed-stake validator model.

- Build a simulation framework** for committee-based PoS systems with validators, staking pools, delegators, reward mechanisms, and attack strategies.
- Model market dynamics** by allowing delegators to react to pool performance and migrate their stake between pools over time.
- Evaluate vote omission and vote delay attacks** under Ethereum and Cosmos reward mechanisms, and compare the results with the fixed-stake analytical baseline.

METHODOLOGY

To study Denial-of-Profit attacks beyond the fixed-stake setting, an agent-based simulation framework was developed. The framework consists of three main actor types: **validators**, **staking pools**, and **delegators**. Validators participate in consensus; a staking pool is represented as a validator that can receive delegated stake from external participants. Therefore, its voting power depends not only on its own stake, but also on the stake delegated to it. Delegators do not participate in consensus directly; instead, they choose a pool and may later migrate based on its observed performance.

The key extension is that validator voting power is no longer fixed. It can change over time because delegators may leave underperforming pools and move their stake elsewhere.

Pool performance is evaluated using two signals:

- APR (annualized percentage return)**: realized return earned by the pool over time
- Reliability score**: how consistently the pool participates in the voting process

These attacks were studied in *Balancing Incentives in Committee-Based Blockchains* by Baloochestani and Jehl. The paper introduced two metrics for evaluating Denial-of-Profit attacks: **effectiveness**, which measures the reward loss inflicted on the victim, and **attack cost**, which measures how much the attacker loses relative to the damage caused.

Using these metrics, the baseline paper analytically showed that vote omission and vote delay attacks are not profitable in the studied Ethereum and Cosmos settings: although the victim loses rewards, the attacker usually loses more. However, this result is obtained in a fixed-stake setting, where each validator's voting power remains unchanged.

Delegators observe these signals and may migrate if their current pool performs worse than alternatives. The migration model uses two complementary paths. The **push path** models dissatisfaction: if a delegator's current pool underperforms for several consecutive rounds, the delegator may leave and select a new pool. The **pull path** models attraction: even if the current pool is not failing, delegators may probabilistically move toward a better-performing pool when the performance gap is large enough.

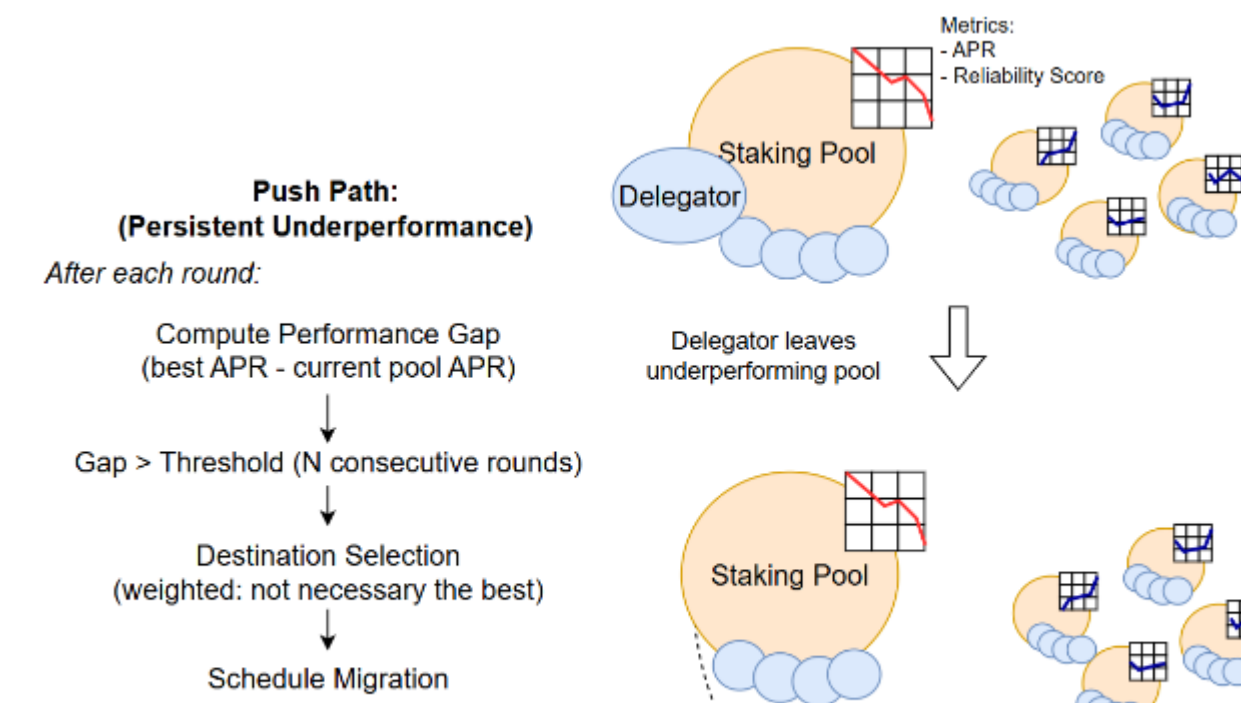


Figure 4: Push-based migration mechanism (delegator leaves a persistently underperforming pool)

The framework is configurable: different reward mechanisms, attack types, attacker/victim powers, and market parameters can be selected. Each experiment is executed twice with the same initial state and random seed: once as a baseline without attack, and once with vote omission or vote delay enabled. The difference between the two runs is then used to calculate attack metrics such as effectiveness and cost.

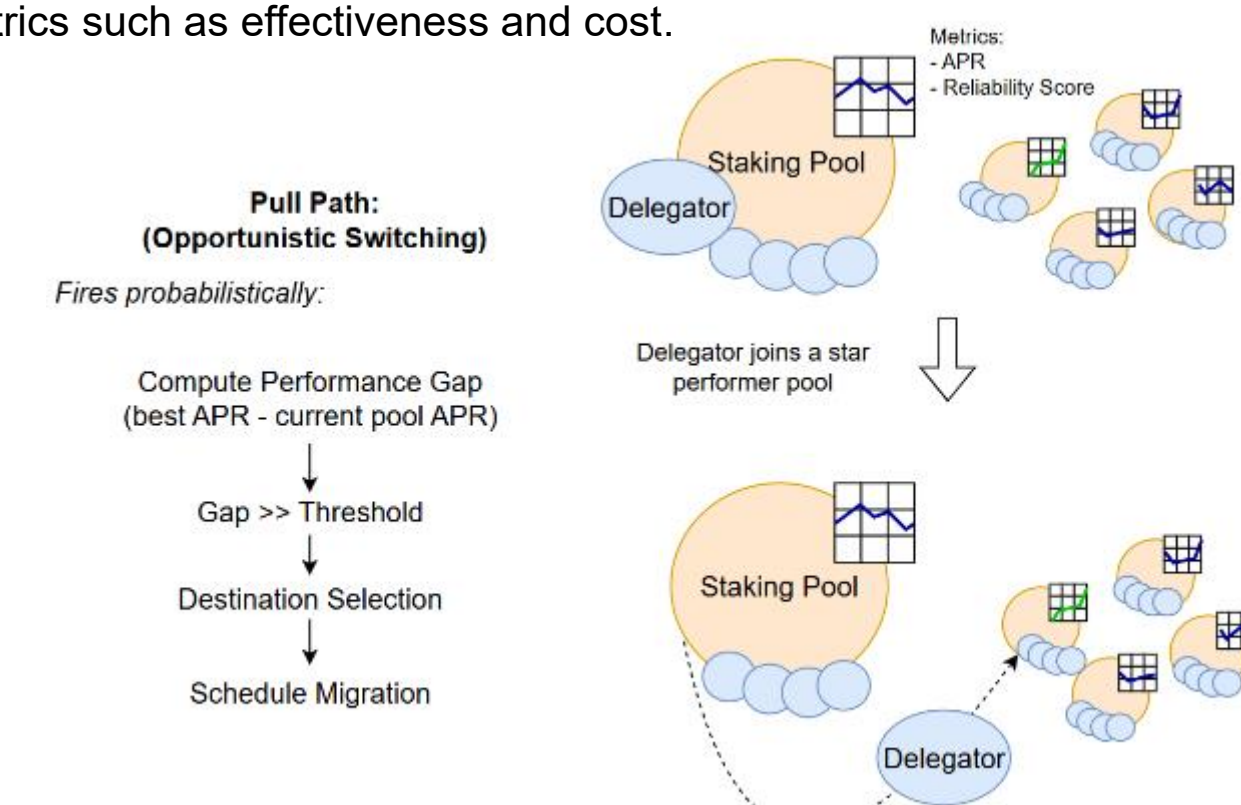


Figure 5: Pull-based migration mechanism (delegator migrates toward a clearly superior pool)

To capture market effects, the original attack cost metric is extended with **cost₂**. While the original cost only compares attacker loss and victim loss, **cost₂** also considers the gain of the best-performing competing pool. This models a scenario where the attacker may be economically connected to, or aligned with, another pool that benefits from delegator migration. In this case, an attack may become more attractive if the attacker's loss is compensated by the competing pool's gain.

RESULTS

The simulation results show that market dynamics can change the interpretation of Denial-of-Profit attacks. In the Ethereum-inspired setting, both **vote omission** and **vote delay** become more damaging when the victim is a small staking pool and the attacker has high voting power. Compared with the fixed-stake analytical baseline, attacks can become **more effective** and **less costly** once delegator migration is enabled.

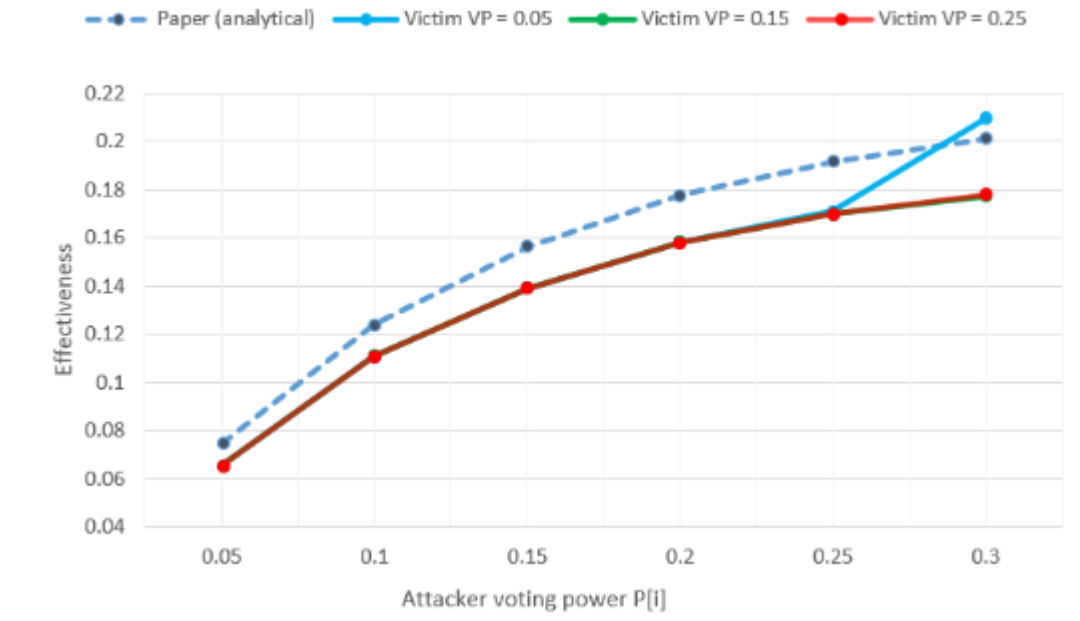


Figure 6: Ethereum. Vote omission attack. Effectiveness. Simulation vs Baloochestani (2025)

The amplification happens because attacks reduce the victim pool's observed performance. Delegators may then migrate to competing pools, causing the victim to lose not only immediate rewards, but also future voting power and market share.

The extended **cost₂** metric shows that market effects can further reduce the effective cost of an attack. In some small-victim scenarios, **cost₂ < 0**, meaning that the gain of a competing or allied pool can outweigh the attacker's direct loss.

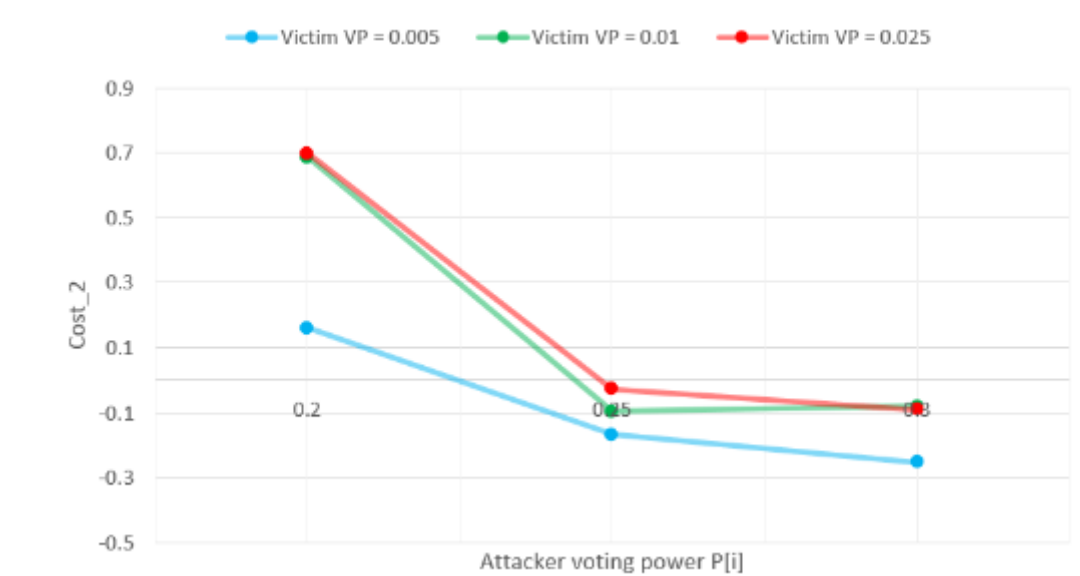


Figure 7: Ethereum. Vote omission attack. Market adjusted Cost. Simulation vs Baloochestani (2025)

In the Cosmos-inspired setting, the simulations remained close to the analytical baseline. The tested reward mechanism did not create large enough APR differences to trigger significant delegator migration, so no strong market-driven amplification was observed.

CONCLUSION

This thesis developed a configurable simulation framework for studying Denial-of-Profit attacks in committee-based Proof-of-Stake blockchains. The framework can model different reward mechanisms, attack strategies, staking pools, delegators, and migration behavior, making it useful for further analysis of blockchain incentive design.

The results show that market dynamics can change the impact of Denial-of-Profit attacks. In some Ethereum configurations, both **vote omission** and **vote delay** attacks become more effective and less costly when delegators are included, especially when the victim is a small pool and the attacker has high voting power.

In contrast, the Cosmos setup stayed close to the analytical baseline. Under the tested parameters, the reward policy kept APR differences too small to trigger significant delegator migration, so pools and delegators did not strongly amplify the attacks.

Overall, the findings suggest that Denial-of-Profit attacks should not be analyzed only in a fixed-stake validator model. **Staking pools, delegator migration, and market dynamics can affect attack severity and should be considered when evaluating Proof-of-Stake reward mechanisms.**